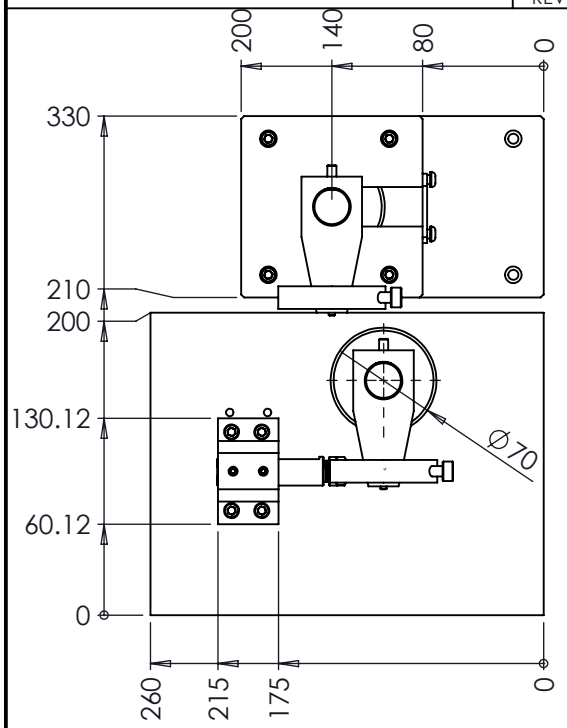
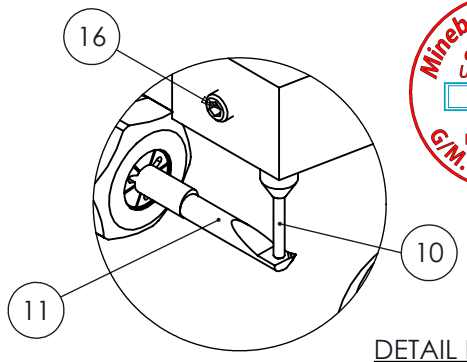
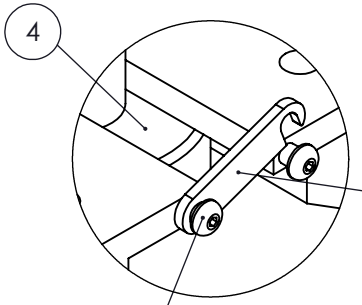
PART No.
TGM-12-384-01A-xx

1 REV.





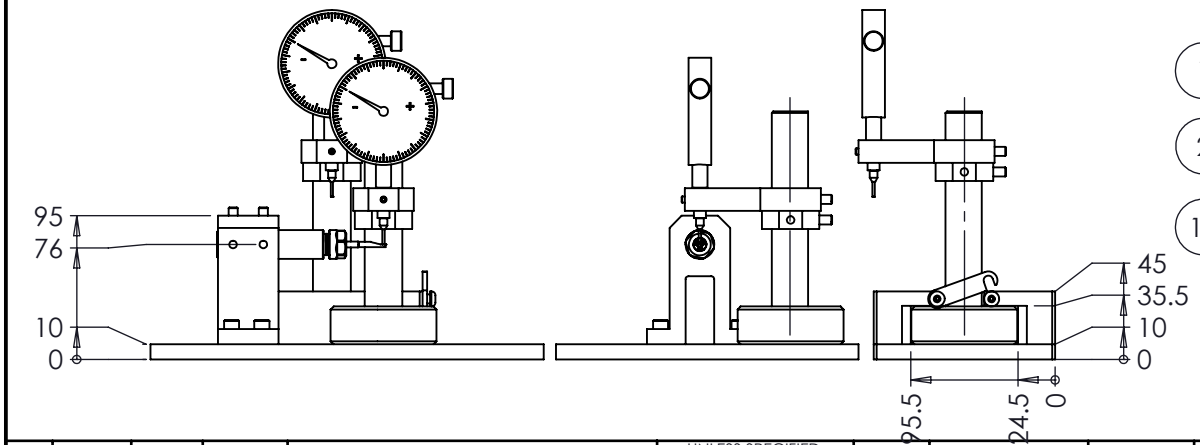
DETAIL D
SCALE 1 : 1

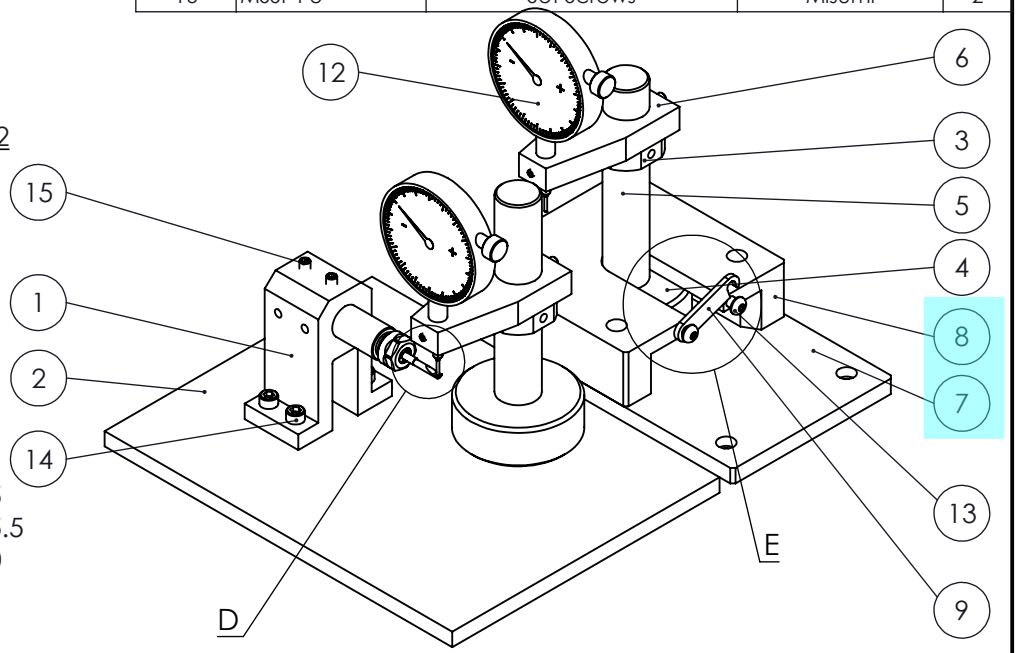


DETAIL E
SCALE 1 : 2

MinebeaMitsumi Inc.
Copy issued Uncontrolled
18/2/2569
Engineering Design center
G/M.1 & M/A. Div.

ITEM NO.	PART NUMBER	PART NAME	DESCRIPTION	QTY.
1	TGM-12-384-1A	Holder center bit (A)	Made to order	1
2	TGM-12-384-2-xx	Base check center boring bit	Made to order	1
3	TGM-12-384-3	Stopper	Made to order	2
4	TGM-12-384-4	Base plate	Made to order	2
5	TGM-12-384-5	Shaft	Made to order	2
6	TGM-12-384-6	Plate for shaft	Made to order	2
7	TGM-12-384-07	Plate lock-1	Made to order	1
8	TGM-12-384-08	Plate lock-2	Made to order	1
9	TGM-12-384-09	Plate lock-3	Made to order	1
10	TGM-02-077B-02	Gauge head	Made to order	2
11	FB-04-S660	Boring bit	PTD.	1
12	Pictest 4M100	Dial Gauge, Pictest	OZAKI MFG	2
13	DBBB4-5-4	Cap Screw	Misumi	2
14	CB6-20	Cap Screw	Misumi	8
15	MSSF6-16	Set Screws	Misumi	6
16	MSSF4-8	Set Screws	Misumi	2





REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S UNCUT

M/C. -
Q'ty 1
MATERIAL See table
FINISH -
HEAT TREATMENT -

List Request No. 269/23
3 RD ANGLE
SCALE : SHEET / UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Supakorn	Chonknok	Preecha	Preecha

DATE 15 Sep' 23

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Jig check center bit

PART No. TGM-12-384-01A-xx

1 REV.

PART No.
TGM-12-384-1A

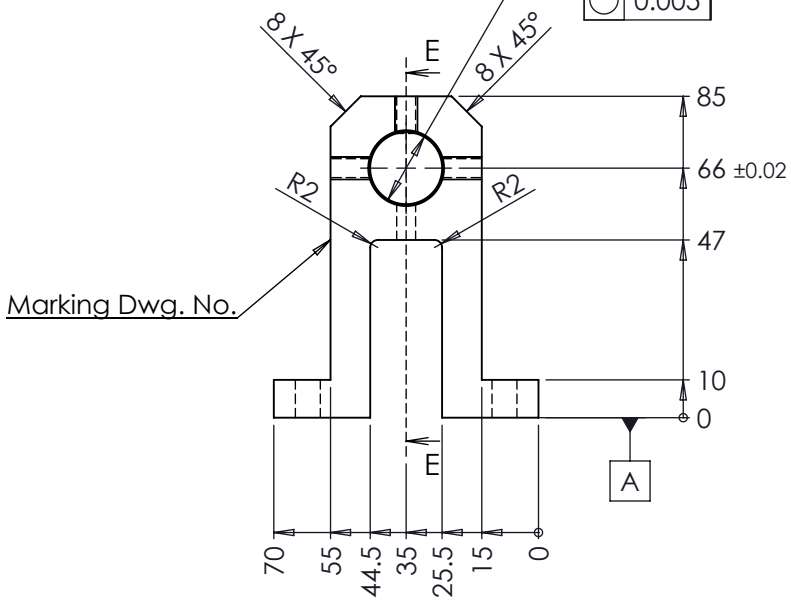
1 REV.

Rt=12 (Rt=6)

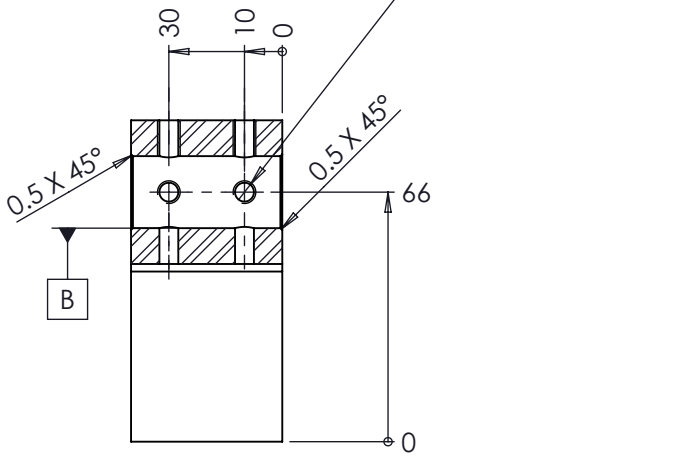
ID. NO BURR
ϕ 19.05 $^{+0.03}_{+0.01}$ THRU ALL

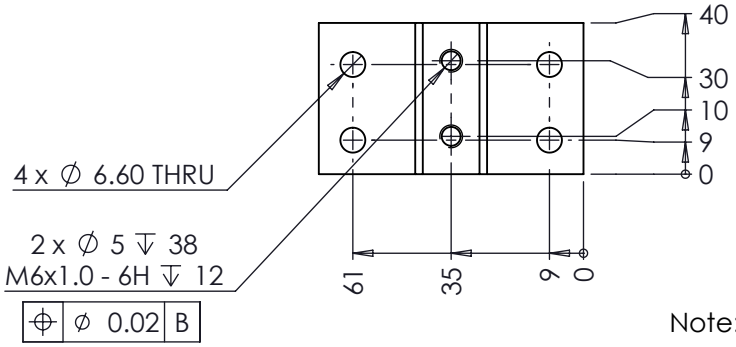
⌖	ϕ 0.01	A
◯	0.005	

1 Pcs

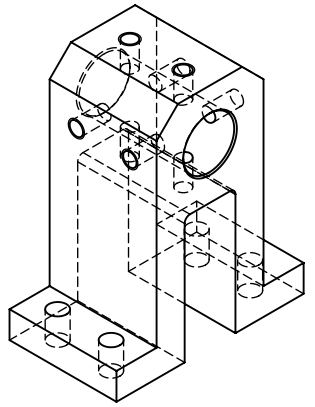


Marking Dwg. No.





SECTION E-E
SCALE 1 : 2



Note: 1. Break all Corner approx by C0.1
2. General Tolerance ±0.1

MinebeaMitsumi Inc.
Copy issued Uncontrolled
18/2/2569
Engineering Design center
G/M.1 & M/A. Div.

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S UNCUT

M/C. -
Q'ty 1
MATERIAL SCM440
FINISH -
HEAT TREATMENT -

List Request No. 347/20
3 RD ANGLE
SCALE : SHEET / UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE 13 Aug' 20

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Holder center bit (A)

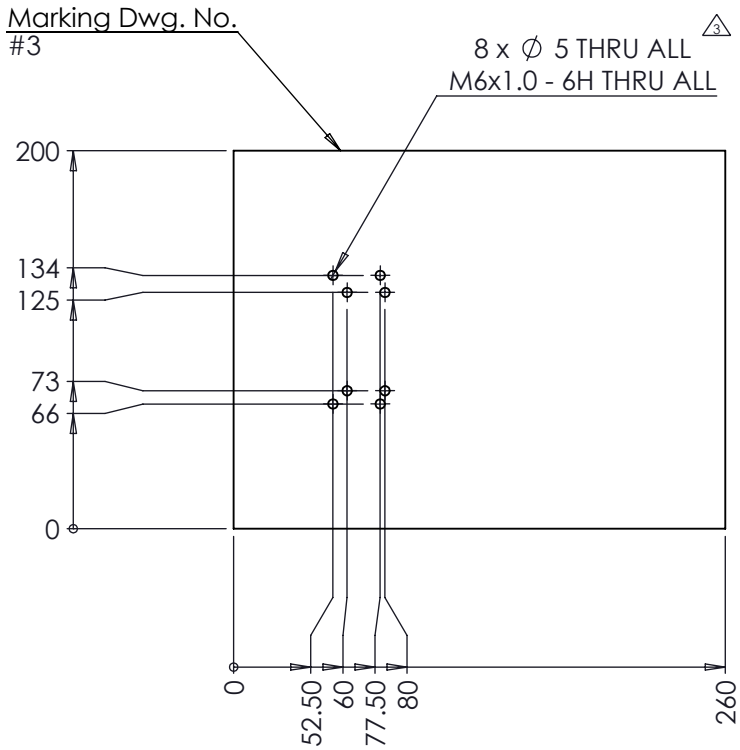
PART No. TGM-12-384-1A

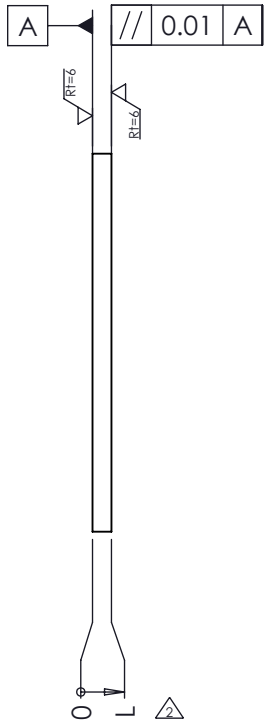
1 REV.

PART No.
TGM-12-384-2-xx

3 REV.

Rt=12 (Rt=6)





Item	L	Check
-01	5	Preecha
-02	10	Preecha
-03	15	Preecha
-04		
-05		

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

MinebeaMitsumi Inc.
Copy issued Uncontrolled
18/2/2569
Engineering Design center
G/M.1 & M/A. Div.

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	13 Aug'20	Preecha	Change: 5 ==> L RQ.347/20
3	----	14 Mar'23	Preecha	Change: M6 (x4) ==> M6 (x8) RQ.68/23

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S UNCUT

M/C. -
Q'ty 1
MATERIAL SUS 304
FINISH -
HEAT TREATMENT -

List Request No. 68/23
3 RD ANGLE
SCALE : SHEET / UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Supakorn	Chonknok	Preecha	Preecha

DATE 10 Feb' 11

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Base check center boring bit

PART No. TGM-12-384-2-xx

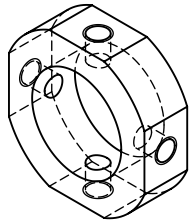
3 REV.

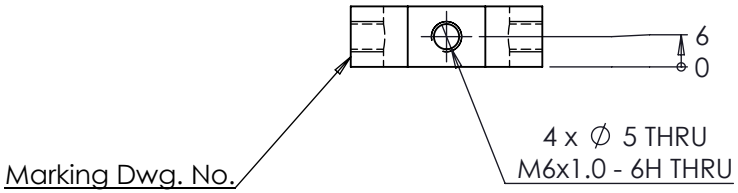
PART No.
TGM-12-384-3

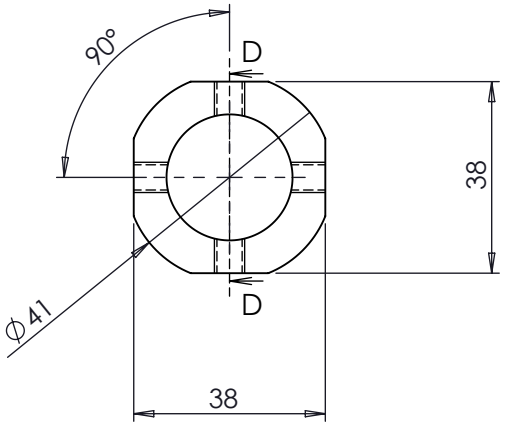
1 REV.

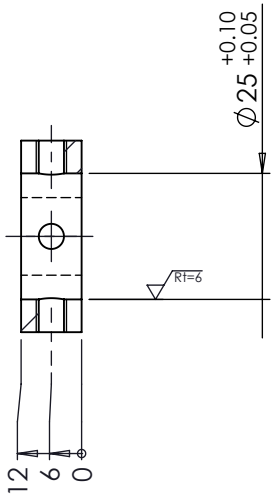
Rt=12 (Rt=6)

1 Pcs









SECTION D-D
SCALE 1 : 1.5

MinebeaMitsumi Inc.
Copy issued Uncontrolled
18/2/2569
Engineering Design center
G/M.1 & M/A. Div.

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S UNCUT

M/C. -
Q'ty 1
MATERIAL SS 400
FINISH Uni Chrome
HEAT TREATMENT -

List Request No. 49/26
3 RD ANGLE
SCALE : SHEET / UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE 25 May' 11

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Stopper

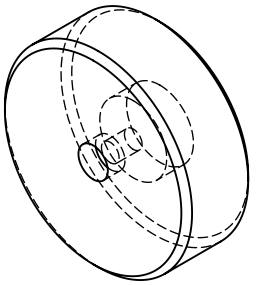
PART No. TGM-12-384-3

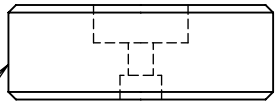
1 REV.

PART No.
TGM-12-384-4

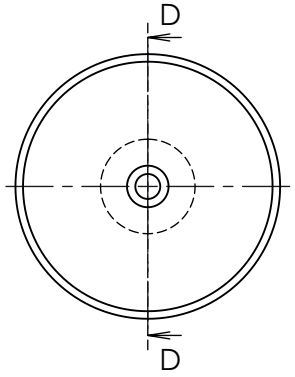
1 REV.

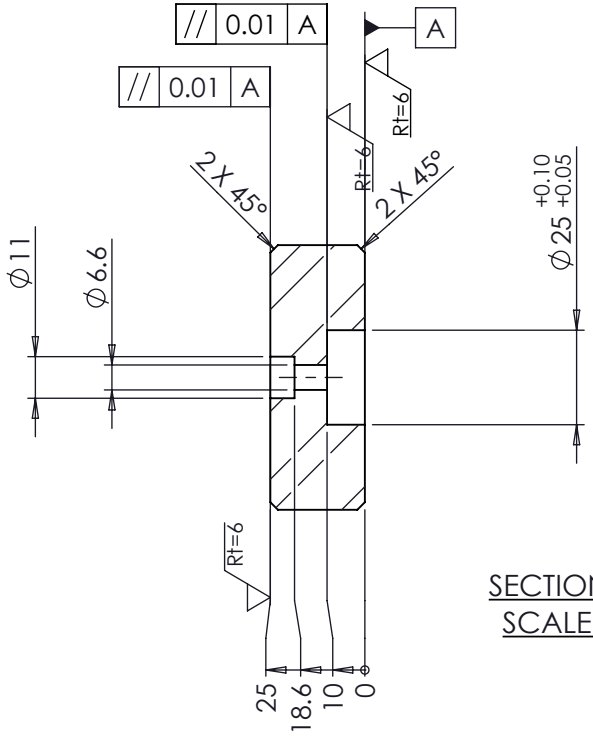
Rt=12 (Rt=6)





Marking Dwg. No.





SECTION D-D
SCALE 1 : 2

MinebeaMitsumi Inc.
Copy issued Uncontrolled
18/2/2569
Engineering Design center
G/M.1 & M/A. Div.

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S UNCUT

M/C. - | Q'ty 1

MATERIAL SS 400

FINISH Uni Chrome

HEAT TREATMENT -

List Request No. 49/26

3 RD ANGLE

SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE 12 May' 11

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Base Plate

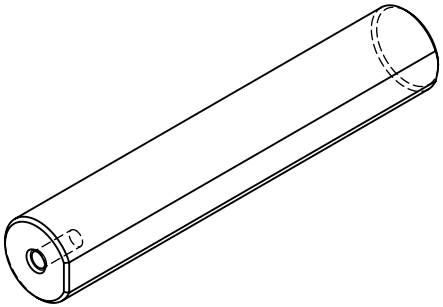
PART No. TGM-12-384-4

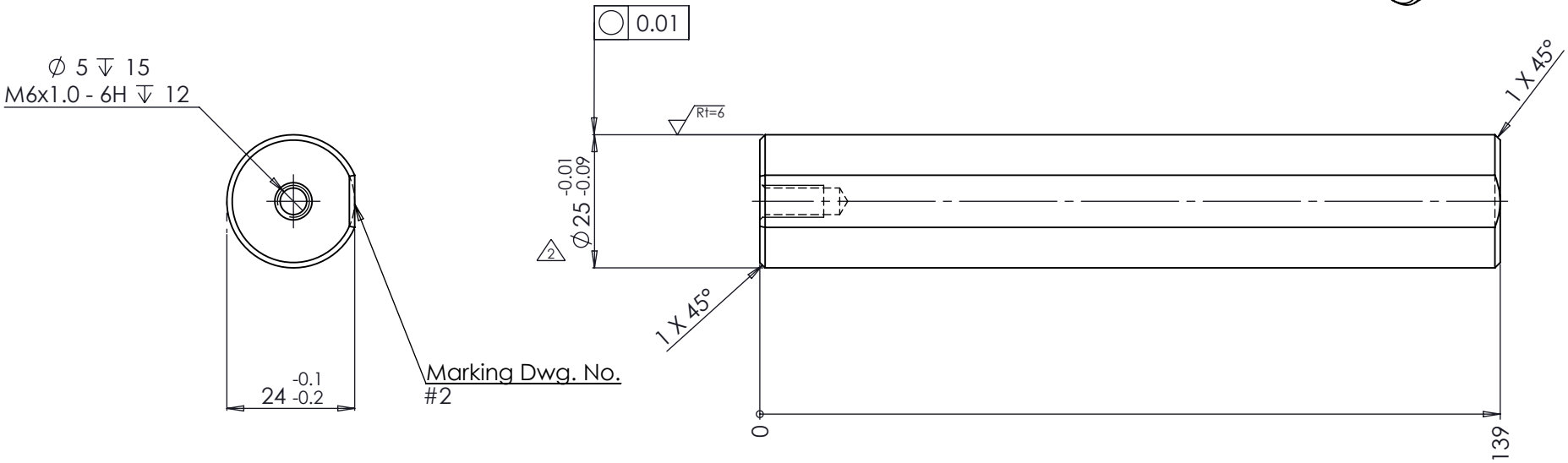
1 REV.

PART No. TGM-12-384-5 | 2 REV.

Rt=12 (Rt=6)

1Pcs





MinebeaMitsumi Inc.
Copy issued Uncontrolled
18/2/2569
Engineering Design center
G/M.1 & M/A. Div.

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	18 Sep'18	Preecha	Add: ∅25 ==> ∅25 -0.01 -0.09

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C.	Q'ty
-	1

MATERIAL: SS 400

FINISH: Uni Chrome

HEAT TREATMENT: -

List Request No. 487/18

3 RD ANGLE

SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE 12 May' 11

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

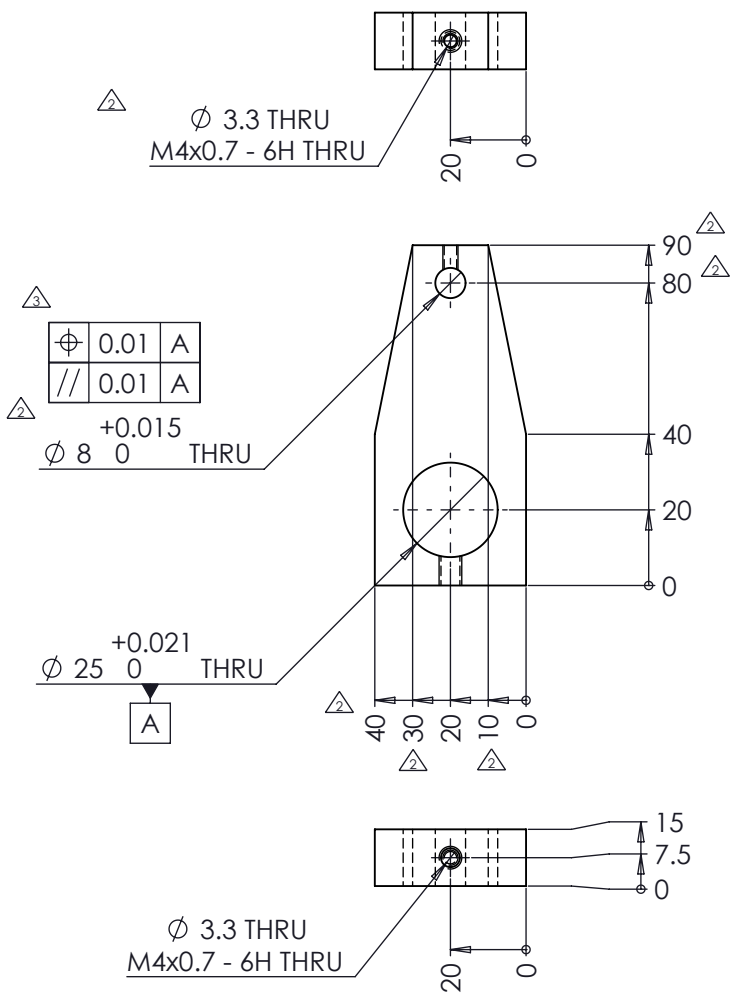
TITLE: Shaft

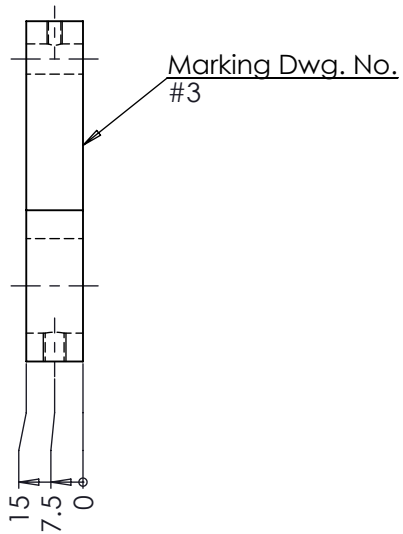
PART No. TGM-12-384-5 | 2 REV.

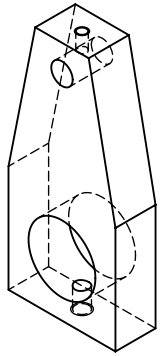
PART No. TGM-12-384-6 | 3 REV.

Rt=12

1Pcs







MinebeaMitsumi Inc.
Copy issued Uncontrolled
18/2/2569
Engineering Design center
G/M.1 & M/A. Div.

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	17 Aug'18	Preecha	Change: 70 ==> 80 etc., RQ.453/18
3	----	18 Feb'26	Preecha	Add: ⌖ 0.01 A // 0.01 A etc.,

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C.	Q'ty
-	1
MATERIAL	
SS 400	
FINISH	
Uni Chrome	
HEAT TREATMENT	
-	

List Request No. -

3 RD ANGLE

SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE 12 May' 11

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

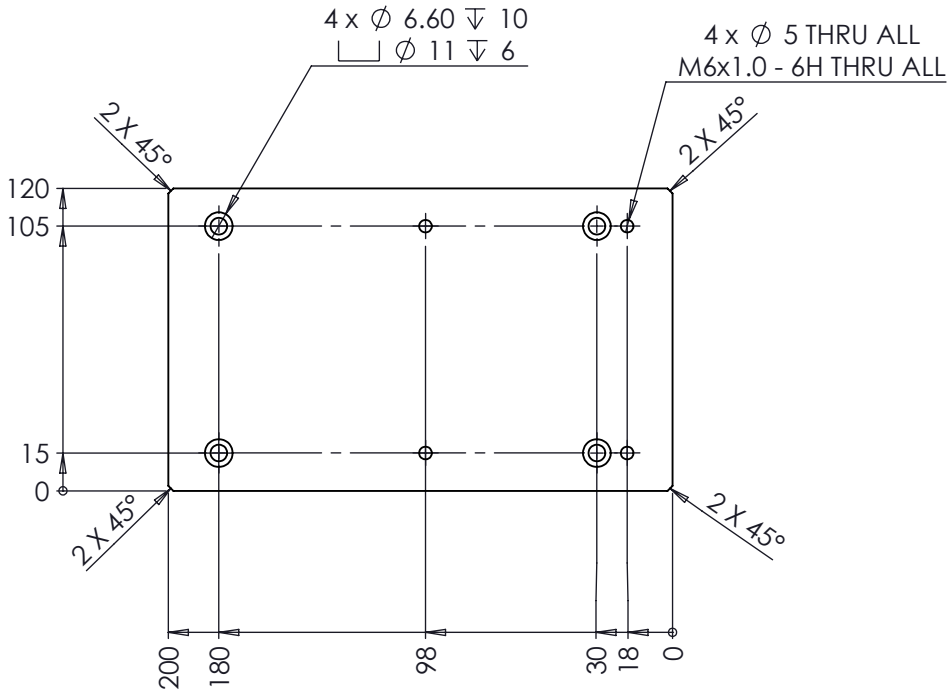
TITLE: Plate for shaft

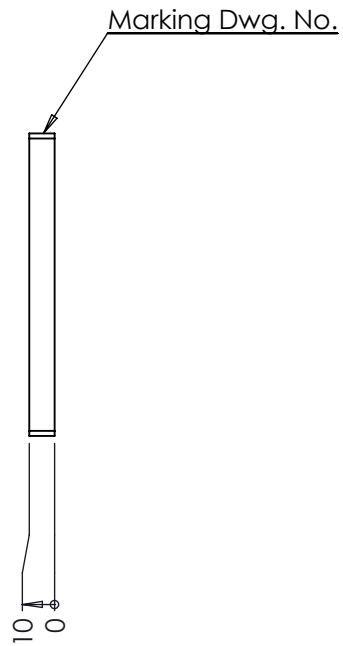
PART No. TGM-12-384-6 | 3 REV.

PART No.
TGM-12-384-07

1 REV.

Rt=6





Note: 1. Break all Corner approx by C0.1
2. General Tolerance ±0.1

MinebeaMitsumi Inc.
Copy issued Uncontrolled
18/2/2569
Engineering
Design center
G/M.1 & M/A. Div.

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S UNCUT

M/C. -
Q'ty 1
MATERIAL SS 400
FINISH Nickel Plating
HEAT TREATMENT -

List Request No. 269/23
3 'RD ANGLE
SCALE : SHEET / UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Supakorn	Chonknok	Preecha	Preecha

DATE 15 Sep' 23

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Plate lock-1

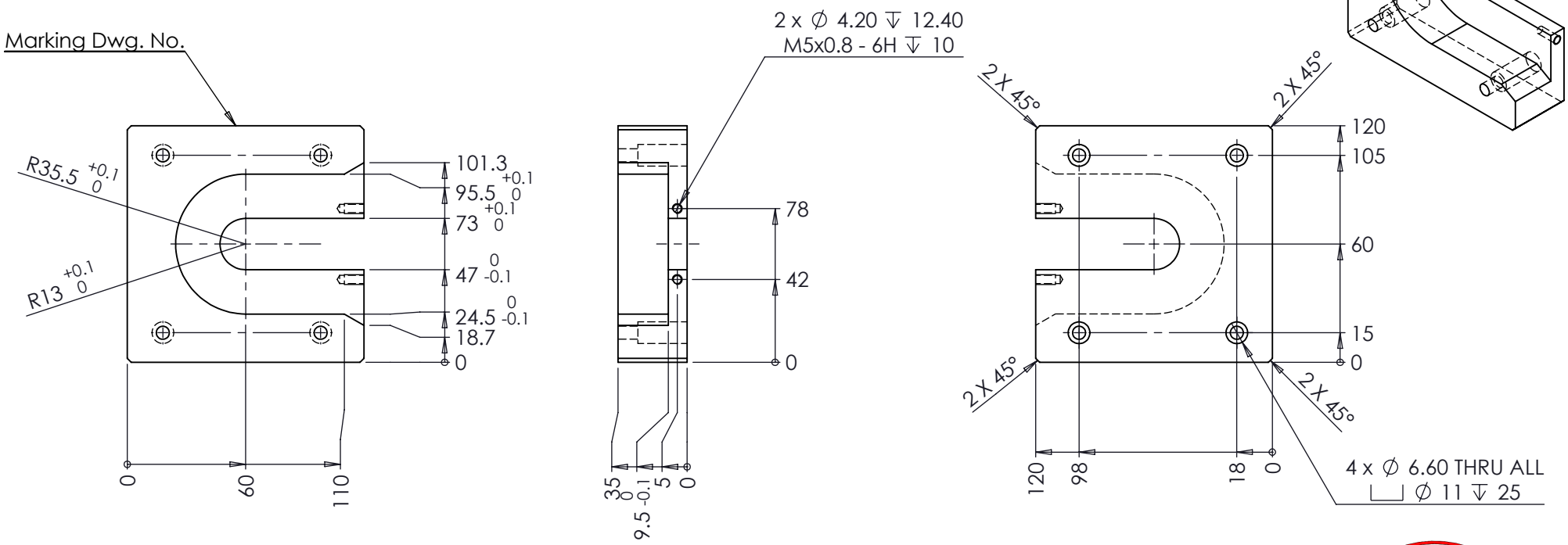
PART No. TGM-12-384-07

1 REV.

PART No.
TGM-12-384-08

1 REV.

Rt=12



Note: 1. Break all Corner approx by C0.1
2. General Tolerance ±0.1

MinebeaMitsumi Inc.
Copy issued Uncontrolled
18/2/2569
Engineering
Design center
G/M.1 & M/A. Div.

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE

UNLESS SPECIFIED C0.1 OR R0.1
GENERAL TOLERANCE
JISB0405 MACHING WORKING

LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S UNCUT

M/C. -
Q'ty 1
MATERIAL PP Gray
FINISH -
HEAT TREATMENT -

List Request No. 269/23
3 RD ANGLE
SCALE : SHEET / UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Supakorn	Chonknok	Preecha	Preecha

DATE 15 Sep' 23

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Plate lock-2

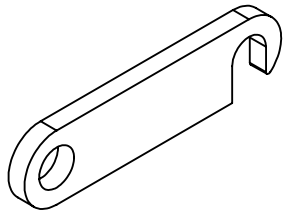
PART No. TGM-12-384-08

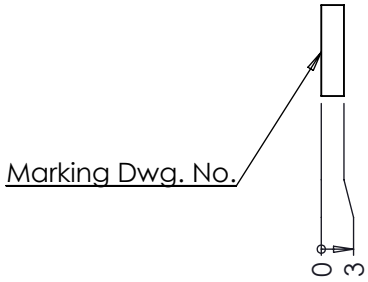
1 REV.

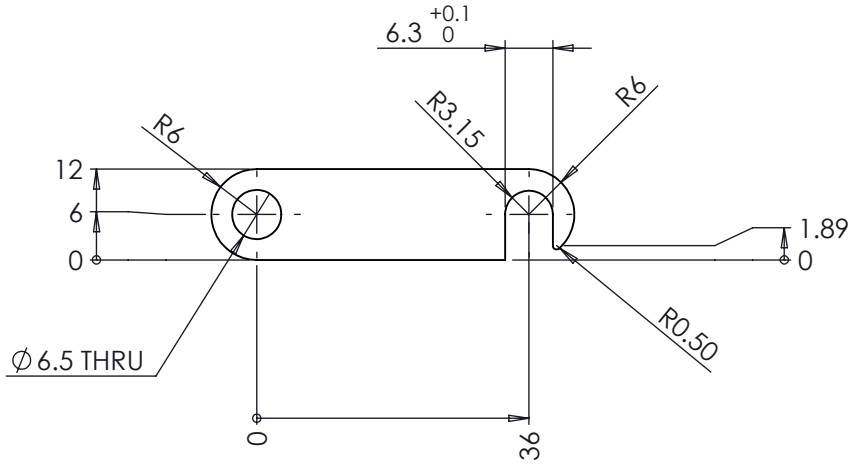
PART No.
TGM-12-384-09

1 REV.

Rt=12







Note: 1. Break all Corner approx by C0.1
2. General Tolerance ±0.1

MinebeaMitsumi Inc.
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Engineering Design center
G/M.1 & M/A. Div.

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S UNCUT

M/C. -
Q'ty 1
MATERIAL SS 400
FINISH Nickel Plating
HEAT TREATMENT -

List Request No. 269/23
3 RD ANGLE
SCALE : SHEET / UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Supakorn	Chonknok	Preecha	Preecha

DATE 15 Sep' 23

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Plate lock-3

PART No. TGM-12-384-09

1 REV.